

Lab 4

Sample code

```
#include <pthread.h>
#include <stdio.h>
#include <stdlib.h>
#include <iostream>
using namespace std;

int sum = 0;           // Shared data

// Thread function
void* runner(void *param)
{
    // Ensure param is not null before converting

    if (param == NULL) pthread_exit(0);

    int upper = atoi((char*)param);

    for (int i = 1; i <= upper; i++) {
        sum += i;
    }

    pthread_exit(0);
}

int main(int argc, char *argv[])
{
    if (argc < 2) {
        cerr << "Usage: " << argv[0] << " <integer_value>" << endl;
        return 1;
    }
    pthread_t threadID;

    pthread_create(&threadID, NULL, runner, argv[1]);

    pthread_join(threadID, NULL);

    cout << "Sum = " << sum << endl;
    return 0;
}
```

```
ubuntu@ubuntu24-04: ~/Desktop/lab4
ubuntu@ubuntu24-04:~/Desktop/lab4$ g++ q1.cpp -o q1 -lpthread
ubuntu@ubuntu24-04:~/Desktop/lab4$ ./q1 10
Sum = 55
ubuntu@ubuntu24-04:~/Desktop/lab4$
```

Q1

```
#include <pthread.h>
#include <stdio.h>
#include <stdlib.h>
#include <iostream>
using namespace std;

void* runner(void *param)
{
    char* message = (char*)param;

    for (int i = 0; i < 5; i++) {
        cout << message << endl;
    }

    pthread_exit(NULL);
}

int main()
{
    char msg1[ ]="Hello";
    char msg2[ ]="World";
    char msg3[ ]="Threads";

    pthread_t thread1,thread2,thread3;

    pthread_create(&thread1, NULL, runner, msg1);
    pthread_create(&thread2, NULL, runner, msg2);
    pthread_create(&thread3, NULL, runner, msg3);

    pthread_join(thread1, NULL);
    pthread_join(thread2, NULL);
    pthread_join(thread3, NULL);

    return 0;
}
```

```
ubuntu@ubuntu24-04:~/Desktop/lab4$ g++ q1.cpp -o q1 -lpthread
ubuntu@ubuntu24-04:~/Desktop/lab4$ ./q1
World
World
World
World
World
Hello
Hello
Hello
Hello
Hello
Threads
Threads
Threads
Threads
Threads
ubuntu@ubuntu24-04:~/Desktop/lab4$
```

Q3

```
#include <pthread.h>
#include <stdio.h>
#include <stdlib.h>
#include <iostream>
using namespace std;

int count;

void* min(void *param) {
    int* array1 = (int*) param;

    int min = array1[0];

    for(int i =0; i<count; i++){
        if(min>array1[i]){
            min = array1[i];
        }
    }
    cout<< "Min: "<< min<<endl;
    return NULL;
}
```

```

void* max(void *param) {
    int* array1 = (int*) param;

    int max = array1[0];

    for(int i =0; i<count; i++){
        if(max<array1[i]){
            max = array1[i];
        }
    }
    cout<< "Max: "<< max<<endl;
    return NULL;
}

void* avg(void *param) {
    int* array1=(int*) param;
    int sum =0;

    for(int i =0; i<count; i++){
        sum = sum+array1[i];
    }

    int avg = sum/count;
    cout<< "Average: "<< avg<<endl;
    return NULL;
}

int main(int argc, char *argv[]) {
    int array[argc-1];

    for(int i =0; i<argc-1; i++) {
        array[i] = atoi(argv[i+1]);
    }

    count = argc-1;

    pthread_t threadID1, threadID2, threadID3;
    pthread_create(&threadID1, NULL, min, array);
    pthread_join(threadID1, NULL);

    pthread_create(&threadID2, NULL, max, array);
    pthread_join(threadID2, NULL);

    pthread_create(&threadID3, NULL, avg, array);
    pthread_join(threadID3, NULL);

    return 0;
}

```

```
ubuntu@ubuntu24-04: ~/Desktop/lab4
ubuntu@ubuntu24-04:~/Desktop/lab4$ g++ q1.cpp -o q1 -lpthread
ubuntu@ubuntu24-04:~/Desktop/lab4$ ./q1 90 81 78 95 79 72 85
Min: 72
Max: 95
Average: 82
ubuntu@ubuntu24-04:~/Desktop/lab4$
```